

CEN314 – Fundamentals of Data Visualization

Final Group Project

Overview

The final group project is a comprehensive, four-week assignment where students will work in teams of 2 or 3 members to apply the concepts and tools learned in the course to a real-world dataset. The project will culminate in the development of an interactive dashboard, a written report, and a 20-minute presentation.

Learning Goals

By completing this project, students will be able to:

- Apply principles of tidy data
- Use appropriate visualizations for different data types and questions
- Create interactive and animated data visualizations using plotly
- Build web-based applications with Shiny
- Communicate insights clearly through written and oral presentations

Requirements

1. Dataset

- Choose a real-world dataset that interests your group. The dataset must:
 - Contain at least 1000 observations
 - Include both categorical and numerical variables
 - Be publicly available

2. Formulate Research Questions

- Formulate 2-3 research questions that are:
 - Relevant to the dataset's context
 - Answerable using visualizations and data summaries
 - Specific enough to guide visualization choices
- Example research questions:
 - What are the trends in CO₂ emissions across continents over the past 50 years?
 - How does internet usage relate to income levels across different world regions?
 - Which factors are most associated with customer churn in a telecommunications dataset?

3. Data Preparation

- Load and clean the data using tidyverse
- Transform the data into tidy format as needed
- Use summaries and joins if multiple data sources are used

- Ensure the cleaning and transformation process is clearly aligned with the research questions
- Justify the selection of variables and any filtering done

4. Static Visualizations

- Create at least five static visualizations using `ggplot2`
- Use at least three different geoms (e.g., `geom_bar()`, `geom_point()`, `geom_line()`)
- Demonstrate use of faceting, color scales, labels, and themes
- Each visualization should be directly tied to one of the research questions
- You should explain how the visualization answers or illuminates the question

5. Interactive Visualizations

- Convert at least two static plots into interactive plots using `plotly`
- Refer to external resources (e.g., plotly-r.com, [geeksforgeeks.org](https://www.geeksforgeeks.org))
- Each visualization should be directly tied to one of the research questions
- You should explain how the visualization answers or illuminates the question

6. Interactive Dashboard

- Develop a basic Shiny app with:
 - A user-friendly UI (inputs, controls)
 - Dynamic outputs (interactive plots, summaries, or tables)

7. Written Report (5–6 pages, double-spaced)

- Begin with a section titled "Research Questions and Rationale"
- Describe the dataset and context
- Explain cleaning and transformation steps
- Interpret key visualizations
- For each visualization or dashboard element, clearly indicate which research question it addresses
- Reflect on limitations and possible extensions

8. Oral Presentation

- Start the presentation by introducing the research questions
- Throughout the talk, connect each component of the analysis back to the questions
- Ensure your presentation lasts 20 minutes
- All team members must participate
- Must include a live demo of the Shiny app

Submission Instructions

Format:

Submit all your files in a single zipped folder via email to the instructor by the deadline.

Folder Naming Convention:

CEN314_FinalProject_GroupX_LastNames.zip

(e.g., CEN314_FinalProject_Group3_Yalcin_Kaya_Demir.zip)

Your zipped folder must include:

1. Report

- Format: .docx
- Includes: Introduction, Research Questions, Data Preparation, Visualizations, Shiny App Description, and Conclusions

2. R Script

- File(s): app.R or a clearly organized ui.R + server.R structure
- All sections (data cleaning, plotting, Shiny UI/server) must be clearly commented
- Include any custom .R scripts if modularized

3. Presentation Slides

- Format: .pptx
- 15–20 minutes of material with slides aligned to the report and Shiny app
- All group members should be represented

4. Data file

- Format: .csv
- Include dataset(s) used in the project

Evaluation Rubric (Total: 70 Points)

Criteria	Points
Data Cleaning & Preparation	10
Use of ggplot2 (static visuals)	10
Interactive Visuals with plotly	10
Dashboard Functionality (Shiny)	15
Report Quality	10
Presentation & Demo	10
Innovation	5

Performance Levels Description

Each rubric criterion is evaluated using four performance levels: Absent, Poor, Acceptable, and Good. The table below describes what each level typically reflects for each criterion.

Data Cleaning & Preparation

Performance Level	Description
Absent	No data cleaning or preparation is evident. Raw data is used without any explanation or transformation.
Poor	Minimal cleaning applied; evident errors or inconsistencies remain in the dataset.
Acceptable	Basic data cleaning and preparation steps are conducted with adequate documentation.
Good	Data is thoroughly cleaned and transformed; explanations and justifications are clear and appropriate.

Use of ggplot2 (static visuals)

Performance Level	Description
Absent	No ggplot2 visualizations are presented.
Poor	Visuals are present but lack coherence, clarity, or are incorrectly constructed.
Acceptable	Visuals demonstrate appropriate use of ggplot2 with some customization and accuracy.
Good	Visuals are well-crafted, accurate, and make strong use of ggplot2 features like themes, labels, and faceting.

Interactive Visuals with plotly

Performance Level	Description
Absent	No interactive visuals are included.
Poor	Interactive plots are minimally functional or unclear in purpose.
Acceptable	At least two interactive plots are functional and relevant to the data.
Good	Interactive visuals enhance understanding of the data and are well-integrated with analysis.

Dashboard Functionality (Shiny)

Performance Level	Description
Absent	No Shiny dashboard is submitted.
Poor	Dashboard exists but lacks core functionality or has major usability issues.
Acceptable	Functional Shiny dashboard with user controls and interactive elements.
Good	Dashboard is fully functional, user-friendly, and includes thoughtful UI design and interactivity.

Report Quality

Performance Level	Description
Absent	No report is submitted.
Poor	Report is incomplete, disorganized, or lacks essential sections.
Acceptable	Report covers all required elements and explains methodology and insights with clarity.
Good	Report is comprehensive, well-written, and clearly communicates the full data story.

Presentation & Demo

Performance Level	Description
Absent	Group does not present.
Poor	Presentation is disjointed, lacks clarity, or fails to cover key project components.
Acceptable	All group members contribute; presentation flows logically and covers main points.
Good	Presentation is engaging, well-organized, and includes a successful live demo.

Innovation

Performance Level	Description
Absent	No additional features or effort beyond minimum requirements.
Poor	Attempts at innovation are underdeveloped or poorly executed.
Acceptable	Some thoughtful additions beyond the core requirements are included.
Good	Project shows creativity and advanced features that enhance user experience.